OFFICIAL SYSTEM DOCUMENTATION

Nexora Mobile Application

Complete Architecture, User Manual & Operational Reference

Platform:	Nexora Personal & Couple Synchronized Suite
Client Architecture:	Flutter 3.x / Dart (Cross-Platform Mobile)
Backend Engine:	ASP.NET Core 10 Web API + SignalR Real-Time Hub
Security & Privacy:	Zero-Knowledge Vault, 3-Tier Granular Visibility, JWT Bearer
Document Version:	Version 1.0 (Phase 1 Engineering Foundation)
Classification:	Private, Secure & Restricted Distribution

EXECUTIVE SUMMARY

Nexora is a private, extensible executive platform designed as a centralized sanctuary for personal organization and intimate couple synchronization. The mobile application delivers seamless real-time collaboration—encompassing notes, shared calendar, encrypted couple chat, a zero-knowledge secret vault, and an AI-powered empathy and communication coach—while strictly upholding sovereign data privacy.

1. Core Architectural & Privacy Foundations

Nexora's engineering foundation is built around unconditional data sovereignty. Unlike standard shared apps, every single entity in the Nexora database adheres to a strict visibility policy enforced on the server:

Visibility Tier	Scope & Accessibility	Security Enforcement
PRIVATE (0)	Accessible exclusively by the resource owner. Strictly hidden from partner, guests, and system administrators.	Enforced via user ID token claim matching at DB query layer.
PARTNER VISIBLE (1)	Visible to the owner and their linked couple partner. Read-only or collaborative as designated.	Verified via active couple relationship lookup in cache/DB.
SHARED / DUO (2)	Co-owned couple asset. Real-time synchronization and bi-directional edit access across both devices.	SignalR channel broadcast restricted to verified couple room.

CRITICAL SECURITY RULE

System administrators have **NO ACCESS** to private notes, couple messages, or vault secrets. Administrative capabilities in Nexora are strictly restricted to system health, feature flag toggles, and audit log compliance monitoring.

2. Authentication & Onboarding Flow

The mobile app manages identity through a secure, non-custodial token storage architecture:

- **Account Registration:** New users register with First Name, Last Name, Email, and strong password. Passwords are salted and hashed on the backend using ASP.NET Core Identity PBKDF2 with 100,000 iterations.
- **Login & Token Issuance:** Upon successful login, the API returns a short-lived JWT access token (15-minute lifetime) and a cryptographically secure refresh token (7-day validity).
- **Secure Persistence:** Tokens are stored in encrypted device storage via `TokenStorage`. On launch, `AuthProvider` automatically validates existing tokens and restores session state silently.
- **Automatic Token Rotation:** When the access token expires, API requests trigger transparent refresh rotation to maintain seamless user experience without interruptions.
- **Instant Logout:** Logging out invalidates the server-side refresh token and immediately purges local caches and tokens.

3. Home Dashboard (The Bento Grid Cockpit)

The primary landing screen utilizes an Apple-style Bento Grid design system, consolidating high-density personal and couple metrics into an intuitive, responsive executive dashboard:

Component	Functionality & Actions
Header Bar	Displays user avatar with real-time green connection pulse, personalized time-of-day greeting (Morning/Afternoon/Evening), universal search button, and instant logout.
Couple Space Hero	Full-width gradient card with milestone counter (*'Day X Together'*), partner avatar link, and direct one-tap shortcut to the real-time E2EE Duo Chat.
Notes Bento Card	Displays total notes count, pinned note title, and one-tap (+) quick note creation button.
Calendar Bento Card	Highlights next upcoming agenda item, privacy level indicator (Private vs Duo), and event time.
AI Compass Card	Contextual assistant shortcuts: *'Draft dinner invite'* and *'Tone check draft'* with one-tap suggestions.
Vault & News Mini-Cards	Vault preview with masked secret indicator (••••••••) and live news ticker highlighting latest curated headlines.

4. Couple Space & Mutual Cryptographic Pairing

Nexora's couple architecture enforces strict 1:1 monogamous pairing. Users cannot accidentally link multiple accounts, ensuring an exclusive private sanctuary. The screen operates in two dynamic states:

State A: Unpaired Setup Workflow

1. **Generate Invite Code:** Partner A taps *'Generate Invite Code'*. The backend creates an alphanumeric single-use token (e.g. `NEX-7K9M2P`) with a strict 48-hour expiration window.
2. **Share Securely:** Partner A taps the copy icon to transfer the invitation code out-of-band via secure messenger or in-person.
3. **Enter Code & Bond:** Partner B inputs the invite key into *'Enter Partner\'s Code'* and taps *'Connect Couple'*. Both accounts instantly transition into a mutually synchronized relationship.

State B: Paired Couple Hub

Once paired, the screen transitions to the active Couple Hub featuring:

- **Celebration Header:** Highlights mutual names and date of connection.
- **Days Together Milestone Counter:** Dynamic calculated live counter tracking relationship duration.
- **Encrypted Link Status:** Live SignalR connection status verifying mutual channel encryption.
- **Dedicated Duo Actions:** Direct launchpads for Couple Chat and Shared Duo Calendar.

5. Personal Notes & Knowledge Capture

The Notes module offers full Markdown authoring, rich metadata tagging, and granular sharing controls:

- **Pinning & Organization:** Pin high-priority notes to top of feed and Home dashboard.
- **Visibility Selector:** Choose whether a note is Private, Partner-Visible (read-only for partner), or Shared (co-authored).
- **Search & Filter Tabs:** Filter by visibility badge (All, Private, Duo) and real-time keyword search.
- **Haptic Feedback & Swipe Actions:** Seamless swipe-to-archive and quick delete interactions.

6. Real-Time Couple Chat (SignalR WebSockets)

The built-in chat engine provides zero-latency messaging between paired partners:

Capability	Technical Specification
Protocol	ASP.NET Core SignalR WebSockets with automatic fallback to Server-Sent Events / Long Polling.

Channel Security	Room IDs are mapped to Couple ID hash; non-members are rejected server-side with HTTP 403.
Message Lifecycle	Persisted in SQL Server with UTC timestamping, soft delete support, and instant delivery receipts.
UI UX Flow	Auto-scrolling message list, bubble alignments (owner on right, partner on left), and timestamp badges.

7. Knowledge Vault (Zero-Exposure Secret Store)

The Knowledge Vault safeguards sensitive items such as passwords, banking recovery keys, identity papers, and emergency pins:

- **Item Categorization:** Classify items under Documents, Passwords, Payment Cards, or Custom Secrets.
- **Masked Previews:** Secret payloads are never shown in plaintext on list views (rendered as ••••••••).
- **One-Tap Decrypt & Copy:** Tap eye icon to unmask value; tap copy to securely copy to clipboard with instant confirmation toast.
- **Duo Shared Secrets:** Mark critical home items (Wi-Fi password, insurance policy) as Shared to make them accessible to your partner.

8. AI Intelligence & Communication Coach

Nexora integrates an AI assistant specifically tuned for personal productivity and relational empathy. The engine operates via provider abstraction (Cloud AI or private local Ollama):

Mode 1: General Assistant (Ask AI)

- Formulate questions, summarize long thoughts, or draft invitations with optional context tags.

Mode 2: Relationship Coach

- **Tone & Empathy Analysis:** Paste a draft message or describe a tense discussion situation.
- **De-escalation Feedback:** The coach identifies emotional subtexts, flags accidental hostility, and offers empathetic phrasing alternatives.
- **Recommended Rephrasing:** One-tap copy of balanced, constructive revisions.

9. Calendar, Curated News & Universal Search

- **Calendar:** Plan personal schedules and joint duo anniversaries. Toggle between agenda list and monthly view with full recurrence support.
- **News Feed:** Aggregates handpicked news articles from verified publishers. Completely free of algorithmic engagement traps or ads.
- **Universal Search:** Global search bar on Home screen queries across notes, calendar entries, vault documents, and messages in a single unified interface.

10. Troubleshooting & Reference FAQ

Question	Answer / Solution
Invite code shows expired or invalid?	Couple invitation codes expire after 48 hours. Generate a fresh code from Partner A's device and enter it within the valid time limit.
Can I use Nexora without a partner?	Yes! Nexora functions as a comprehensive personal productivity system. Couple features remain dormant until an invitation is paired.
How is my data backed up?	All records are synchronized to your self-hosted or dedicated ASP.NET Core database backend with automated transaction safety.
Can my partner see my private vault?	No. Items in the vault default to Private (Visibility 0) and remain visible only on your authenticated account unless explicitly flagged as Shared.